Artooka Vendor Telegram BOT Management Plan

[Artooka]

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1. Document Version

Version	Description	Date	Author(s)
1	Initial project and scope plan	27/09/2026	Jason Feng
2	*Project proposal*	*10/9/2026*	*Hiroshi Inaoka* *Jaeyeong seol*
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2. Introduction

2.1. Team contact information

Name	*Role*	*Email*
Jason Feng	*Dev*	*jfeng56@my.bcit.ca*
Hiroshi Inaoka	*Dev*	*hinaoka@my.bcit.ca*
Mursib Abbasi	*Dev*	*mabbasi12@my.bcit.ca*
Shubh Hanspal	*Scrum master*	*shanspal1@my.bcit.ca*
Vincent Kim	*Dev*	*vkim16@my.bcit.ca*
Jaeyoung Seol	*Dev*	*jseol3@my.bcit.ca*
Johnny Zhang	*Supervisor*	*Johnny.zhang@bcit.ca*

2.2 Stakeholders and target group

- *Student Project Team: Designs, develops, tests and documents the bot. The team needs clear requirements, access to the development environment and regular feedback from Artooka. Contact: See Section 2.1*
- *Supervisor: Provides academic guidance and reviews of project progress and deliverables. Contact: Johnny Zhang, johnny_zhang@bcit.ca*
- *Artooka Sponsor: Define business requirements, prioritizes features and reviews the completed work. They need a bot that helps vendors receive order notifications and manage daily tasks through Telegram. Contact: Pouya Mehrannia, pouya@artooka.com*
- *Artooka Technical Mentor: Provide technical guidance and review backend integration, security and deployment. They need a maintainable solution that integrates with Artooka's existing system without disrupting current services. Contact: Iman Anoosheh, Iman@Artooka.com*
- *Artooka vendors(Primary Target User): Approved vendors will use the bot to receive order alerts, view order details, update delivery status and manage products. They need a simple reliable mobile workflow in Persian and English. Contact: Pilot vendors to be identified through the Artooka sponsor.*

3. Goal and Vision

3.1. Project problem statement

Artooka vendors currently receive new order notifications only by email, which they may not check regularly. This can delay order processing and put timely delivery at risk, especially for flowers and gifts that must arrive on a specific date. Vendor also need to access the web portal to view order details. Update delivery status and manage products. Since Telegram is widely used by Artooka's target vendors, there is an opportunity to provide timely notifications and easier access to common task through a platform they already use.

3.2. Project goal and value

What is the client trying to achieve? What value is this product delivering to the stakeholders?

The goal of this project is to develop a Telegram bot that enables approved Artooka vendors to securely link their accounts, receive new order notifications, view order details and update delivery status. These core features will support Persian and English and integrate with Artooka's existing backend.

3.3. Existing System

Artooka currently provides a web-based vendor portal where vendors can view orders, update delivery status, manage products and check sales and payment information. The portal connects to an existing Django REST Framework backend hosted on Azure App Service, with PostgreSQL used for data storage.

The backend already provides APIs for vendor operations and sends email notifications when new orders are paid. However, it does not currently provide Telegram order notifications or a Telegram interface for vendor tasks. Our project will reuse the existing vendor APIs and add the integration needed for the core Telegram bot features.

3.4. Alternative/Competitor Solutions

Possible alternatives include continuing to use the existing vendor web portal and email notifications, adding browser push notifications or developing a dedicated mobile application. Continuing with the current system would require little development but would not address delayed awareness of new orders. Browser notifications development and maintenance work.

The sponsor has proposed Telegram because it is already widely used by the target vendors and can provide access to essential order tasks from their phones. Our proposed solution focuses on core order management features and reuses the existing backend APIs to keep the project scope manageable. No specific competing products have been identified in the supplied project materials.

4. Proposed Solution

4.1. Proposed SolutionProposed Solution

We propose to develop a Telegram bot that allows approved Artooka vendors to perform essential order management task from their phones. The core scope includes secure account linking and unlinking, new order notifications, viewing order details and updating delivery status. The bot will support Persian and English

The bot will develop in Python using aiogram, with httpx used to communicate with Artooka's existing REST APIs. It will run as a separate service on Azure App Service. PostgreSQL will store conversation state, while a small Django integration will handle account linking and bot authentication. Vendor will link their accounts using an email verification code without entering their passwords in Telegram.

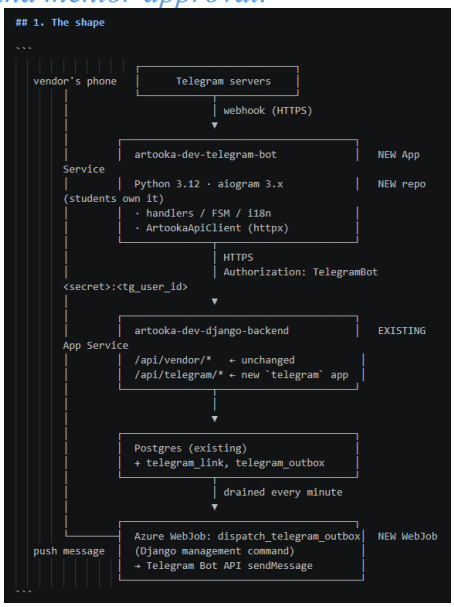
Order notifications will initially use periodic checks of the existing API. Event-driven notifications may be added if time allows and the technical mentor approves the required backend changes. Existing backend services will continue to manage order data and delivery confirmation emails.

The intended deliverable is a working, tested bot suitable for a limited vendor pilot, together with source code, setup instructions and basic maintenance documentation. Production deployment will depend on Artooka's review and approval. Product management, AI-assisted listing and sales summaries will remain stretch goals, subject to sponsor agreement.

4.2. Diagrams/ Analyses

4.3. Client Diagrams

The diagram represents the sponsor's broader architecture. For our proposed core scope, order notifications will initially use polling. The event-driven outbox and dispatcher shown in the diagram are an optional enhancement, subject to available time and mentor approval.



4.4. High-level key features

- ***Secure an account linking and unlinking****: Approved vendors can connect heir Artooka accounts to Telegram using an email verification code and disconnect when needed.*
- ***New order notifications:*** *The bot periodically checks for new orders and sends alerts to the linked vendor through Telegram.*
- ***Order viewing:*** *Vendor can view recent orders and open order details, including items, quantities, recipient information, delivery address, delivery date, and gift message.*
- ***Delivery status updates:*** *Vendors can mark order items as delivered after confirmation. Updates are saved through the existing backend and use its customer email notification process.*
- ***Persian and English Support:*** *the bot provides message buttons and error guidance in both languages.*

4.5. High-level Product Roadmap

What is the order that the team will follow for implementing the above-mentioned features? Does the client have a preference as to how you go about your work? If so, you should explain how you will meet their expectations.

Connect the goals, tasks, features and deliverables with your roadmap.

This should be high-level as you will have a chance to iterate through this as you proceed.

Milestone 2: Development foundation	• *Confirm the core scope with the sponsor* • *Set up the development environment and deployment pipeline* • *Implement basic Persian and English bot responses.* • *Implement secure account linking and unlinking* • *Connect the bot to the existing vendor for APIs.*
Milestone 3: Core development and testing	• *Implement recent order viewing and order detail messages.* • *Implement delivery status updates* • *Implement polling-based new order notifications.* • *Test vendor access permissions and error handling.* • *Demonstrate core workflows and collect sponsor feedback.*
Milestone 4: Testing and delivery	• *Complete integration and user acceptance testing.* • *Fix defects and verify Persian and English workflows.* • *Prepare setup instructions and maintenance documentation.* • *Demonstrate the completed bot and hand over the source code.* • *Conduct a limited vendor pilot, subject to sponsor arrangements.* • *Deploy to production if approved by Artooka.*

4.6. Deliverables

Deliverable	Description
Working Telegram Bot	*A tested bot supporting account linking and unlinking, polling-based order notifications, order viewing and delivery status updates in Persian and English.*
Source Code	*The bot source code and required backend integration code, including dependency files and configuration examples without secrets.*
Setup and Maintenance Documentations	*Instructions for setup, configurations, deployment and basic troubleshooting, include how to disable the bot and manage its credentials*
User Guide	*Brief instructions explaining how vendors can link their accounts and use the core bot features.*

Test Result	*Test cases and results covering core workflows, vendor access permissions, error handling and bilingual support, together with any know limitations.*
Final Demonstration and Handover	*A demonstration of the completed core features and a handover of the code, documentation and remaining backlog to Artooka.*

5. Key Assumptions Assumptions

5.1. Non-Technical Assumptions

- *The sponsor will review and confirm the proposed core scope, with additional features treated as stretch goals.*
- *The sponsor will provide a main contact t clarify requirements and give regular feedback.*
- *The team will have access to a technical mentor to review integration work and help resolve blockers.*
- *Artooka will help arrange vendor feedback or suitable internal testers for acceptance testing.*
- *Team members will be available to complete their assigned work within the agreed course schedule.*
- *Artooka will confirm responsibility for hosting and service costs, production approval and maintenance after handover.*

5.2. Technical Assumptions

- *Artooka will provide access to the development backend, relevant repositories, and the Azure resources needed for the project.*
- *A development vendor account with sample orders and product will be available for integration testing.*
- *The existing vendor APIs will support order viewing and delivery status updates without major changes.*
- *Artooka's existing email service will support account verification codes and delivery confirmation emails.*
- *Python, aiogram and PostgreSQL will be used for the bot service and its persistent state.*
- *Polling the existing order of API will be sufficient for the initial pilot, with the polling interval agreed with the technical mentor.*
- *Required backend changes will be reviewed by the technical mentor and validated through Artooka's CI and development environment.*
- *Development and production will use separate Telegram bot tokens and secrets with credentials kept outside the source code.*

6. Risks and Issues and Issues

Risk or Issue Description	**Potential Impact**	**Mitigation/Resolution Steps**
Development access or realistic test data is delayed	*API integration and testing may be blocked*	*Request access and test accounts early. Use mocked API responses while waiting and escalate blocker to the mentor.*
Backend changes or database migrations fail.	*Existing services may be disrupted or project progress delayed*	*Keep changes small, obtain mentor review, validate in the development*

		environment and document rollback steps.
Account linking or access control are implemented incorrectly	*Vendor could access another vendor's order information or perform unauthorized updates*	*User backend authentication and ownership checks. Test invalid unlinked and cross-vendor requests and obtain mentor review.*
Additional requests or limited team capacity increase the workload.	*Core features and testing may not be completed on time*	*Prioritize the agreed core scope, review progress weekly and defer stretch goals until core testing is complete.*

7. Signatures

7.1. Students

Hiroshi Inaoka
Jaeyeong Seol
Shubh Hanspal
Jason Feng
Mursib Abbasi
Vincent Kim

7.2. Sponsor(s)

Approved

Pouya Mehrannia
Oct 14, 2026

